Predicting Antimicrobial Resistance Without Local Data: Cross-Country Transfer to Kazakhstan

Abstract

Background. Machine-learning models predict antimicrobial resistance (AMR) well in high-income countries but cannot be deployed where no local surveillance exists. With 1.14 million deaths attributed to AMR in 2021, countries such as Kazakhstan lack the isolate-level data needed to guide empirical treatment. We ask whether AMR can be predicted for a country contributing no training isolates, using only publicly available macroeconomic and antibiotic-consumption indicators plus organism and antibiotic identity.

Methods. A CatBoost model was trained on ∼11.3 million susceptibility results from ∼1.08 million isolates across 85 countries, with Kazakhstan excluded entirely. Resistance probabilities were estimated via a leave-one-country-out (LOCO) model and validated by temporal hold-out and four-country hold-out, with AUC, recall, and expected calibration error (ECE) as metrics.

Results. On the post-2020 hold-out, the model achieved an AUC of 0.849, recall of 0.797, and ECE of 0.012 after isotonic calibration. Across four held-out countries (prevalence 15–44%), discrimination generalised (AUC 0.81–0.89) while calibration degraded with dissimilarity. In Kazakhstan, 66.9% of 480 organism–antibiotic–year inferences exceeded the 20% threshold, with *Acinetobacter baumannii* highest.

Conclusions. Macro-feature transfer can prioritise AMR testing in data-poor settings, flagging which combinations to test first — hypothesis-generating, not a surveillance substitute.

Introduction

The public health impact of antimicrobial resistance (AMR) is significant, with antimicrobial resistance being responsible for 4.71 million deaths and 1.14 million direct AMR deaths in 2021[1]. If the trend continues, the global number of AMR cases will hit 10 million annually by 2050[2], while global losses of annual GDP by 2030 could be as high as 3.4 trillion USD[3]. Although AMR is a worldwide challenge, the severity of the impact is not uniform and is more apparent in LMICs as a result of a lack of suitable laboratory diagnostics and infection control capacities[4]. Isolate data from global surveillance databases such as WHO GLASS, ATLAS, and SIDERO-WT are predominantly from high-income countries, and resistance from LMICs is well under-represented. Central Asia, including Kazakhstan, remains a blind spot in global AMR monitoring: post-Soviet states have limited microbiology laboratory capacity and underdeveloped antimicrobial surveillance systems, and no nationwide isolate-based AMR surveillance program exists in Kazakhstan[5]. To close this gap in surveillance, we examine the ability of a model trained exclusively on global data to predict resistance in a country that provided no data for training or validation, which we term the cross-country generalization scenario, referring to a model applied to Kazakhstan with no data from Kazakhstan used to build or train it.

Phenotypic methods for predicting AMR have been shown to work in the absence of genomic data, as do epidemiological and pharmaco-epidemiological predictors[6]. Existing validation techniques, however, assume that isolates from the area of study are available, which is not the case for, for instance, Kazakhstan, as it has limited data availability. Where no local susceptibility data exist, minimizing false negatives is most important: labelling a resistant isolate as susceptible leads to ineffective empirical therapy and an uncleared infection, a more dangerous error than a false positive. The surveillance gap is tackled by applying a CatBoost model trained on information from around the world and tested with only information regarding publicly available macroeconomic indicators and antibiotic consumption as input features.

First, to our knowledge, this is the first application of data-free, cross-country AMR prediction to a Central Asian country (Kazakhstan). We present here three non-local ground truth validation methods, namely, temporal splitting, Leave-One-Country-Out (LOCO), and rolling temporal cross-validation (rTCV). Second, we demonstrate that data readily available could be used to prioritize AMR testing in data-poor settings, yielding a hypothesis-generating approach transferable to other under-surveilled regions. In Kazakhstan, there is no isolate-based system; the main principle of the system is the concept of risk prioritization and to alert which organism–antibiotic combinations should be tested first. The resistance probability estimates based on a model trained with global surveillance data and applied to Kazakhstan restricted to only macro-level data and antibiotic consumption would be sufficient to differentiate antibiotics by high or low risk for prioritization according to the 20 percent clinical actionability threshold.

Methods

2.1 Study Design

A cross-country prediction approach to forecasting AMR patterns for Kazakhstan without the need for local data to be used for model training is introduced in this paper. The proposed pipeline is shown in Figure 1. Two input datasets are used — global AMR surveillance and macro-economic and antibiotic consumption at the national level. Firstly, the global AMR dataset is preprocessed, harmonized, and feature engineered; next, four machine learning models, including LightGBM, CatBoost, Random Forest, and Logistic Regression, are trained to determine the most suitable model among the candidates, excluding Kazakhstan from the sample. Once selected, this model will be transferred to the Kazakhstan domain using only macroeconomic and antibiotic consumption features as inputs. Three robustness measures are applied: hold-out, leave-one-country-out (LOCO) CV, and rolling-origin CV; all three are possible to apply to the data with no local ground truth. Once resistance probabilities are calibrated by the model chosen (with isotonic regression trained on the period of 2019–2020), multicriteria prioritisation leads to the final risk forecast.

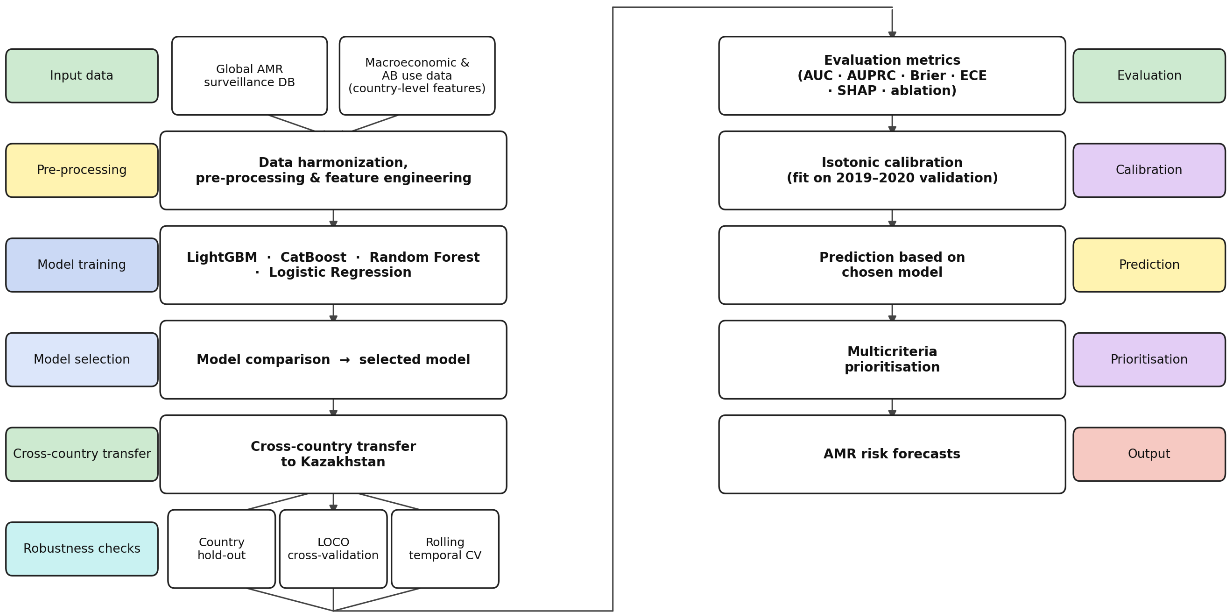

Figure 1: Overall workflow of the machine learning pipeline from data preprocessing to prediction.

2.2 Data

2.2.1 Data Sources and Acquisition

Two types of datasets were used: a global dataset used for all model training, validation, and hyperparameter tuning, and a target dataset for Kazakhstan, where inference and post-hoc face validity were conducted and did not affect any model fitting process. Isolate-level data were obtained from three global surveillance databases accessed via Vivli.org: ATLAS (Pfizer), SIDERO-WT (Shionogi), and KEYSTONE, following applications for data access. The data on consumption of antibiotics (DDD per 1,000 inhabitants per day) were obtained from Wang et al. (2025)[10]. Kazakhstan has no national AMR surveillance system, thus no isolate-level susceptibility data were available for model training. All features were country-level proxy features in the same feature space as the training dataset for the Kazakhstan target dataset. The national average antibiotic consumption rates from four published articles[11,7,12,13] were used. The Kazakhstan feature grid was completed with six macroeconomic indicators (GDP per capita, health expenditure (% of GDP), population density, access to basic water, access to sanitation, and hospital beds per 1,000), annual mean surface temperature (all from World Bank Open Data, 2017–2023)[14], and WHO AWaRe Access percentages[11,16]. The features were only evaluated for their ability to produce resistance probability estimates and were not used in training or calibration of models; only published Kazakhstani resistance rates were evaluated for face validity, and they were never used in model training or calibration.

2.2.2 Data Harmonization

Standard names were used for column names to ensure consistency, and inconsistencies in country names were resolved using a mapping table. Codes that were assigned to an antibiotic were discarded if the code was incorrectly entered as the name of an antibiotic. One of two forms of susceptibility classification was applied, depending on the source database. ATLAS provided pre-computed susceptible (S)/intermediate (I)/resistant (R) interpretations, which were applied directly and accounted for 95.1% of the total labelled results. SIDERO-WT and KEYSTONE reported raw minimum inhibitory concentration (MIC) data, which were binarized using organism-specific clinical breakpoints (EUCAST Clinical Breakpoints v13.0 (2023)[17] where available, and CLSI M100-Ed33 (2023) for organism–antibiotic combinations not covered by EUCAST;. The remaining 4.9% of results were derived from MIC-based labels. Intermediate isolates were dropped to ensure a clear division between susceptible and resistant isolates. The names of antibiotics in the Kazakhstan dataset were converted to lowercase, and partial matching was performed with the global set of antibiotics; antibiotics that could not be matched were excluded from the dataset.

2.2.3 Feature Engineering

The AMR data were combined with macroeconomic and consumption features into a yearly country-level dataset. Features showing right skewness, where most countries have moderate values but a few wealthy countries have very high values (such as GDP and consumption data), were logarithmised using $\log(1+x)$. For Kazakhstan, no population density data were available from the World Bank; therefore, this was calculated manually using population and surface area data. Chronological forward-filling was applied to missing values, and the global mean for each feature over the training period (prior to 2018) was used for any remaining missing values, so that no validation or test data influenced the filling. The final feature set comprised 13 features: 3 categorical (organism, antibiotic, antibiotic class) and 10 numeric (year, 9 macroeconomic features). When applied to the entire training set, forward-filling resolved most missing macro-feature values; global mean imputation was required for a median of 9.4% of country-year observations per feature, with the highest missingness in aware access pct (86.2%, 53 countries) and consumption DDD per class (57.9%, 52 countries). Core World Bank indicators (GDP per capita, population density, health expenditure) had the lowest missingness (7.9%), reflecting the fewest countries lacking World Bank reporting coverage. All features are listed in Table 1.

Table 1: Final feature set used for model training and inference.

Feature Type	Features
Categorical	organism, antibiotic, antibiotic_class
Numerical	year, log_gdp_per_capita, health_expenditure_pct_gdp, population_density_per_sq_km, log_consumption_ddd_per_class, sanitation_access_pct, water_access_pct, hospital_beds_per_1000, mean_temp_celsius, aware_access_pct

2.3 Cross-Country Generalization Design

The model was not trained, validated, or subjected to early stopping on data from Kazakhstan. It was constructed based on global AMR data and applied to Kazakhstan as an unseen region, assuming that no information about AMR status in Kazakhstan itself was known, but only proxy variables related to macroeconomic indicators and antibiotic consumption. National antibiotic consumption data (defined daily doses per 1,000 inhabitants per day, 2017–2023) were combined with the six priority pathogens identified by WHO (*E. coli*, *K. pneumoniae*, *S. aureus*, *P. aeruginosa*, *A. baumannii*, *E. faecium*) to produce 480 organism–antibiotic–year inferences (18 antibiotics $\times$ 6 pathogens $\times$ 7 years) for Kazakhstan's inference grid. All features were imputed and log-transformed as was done with the worldwide dataset (Sections 2.2.2–2.2.3), though the pharmacological class was also constructed.

The first model implemented was CatBoost with the leave-one-country-out (LOCO) method, where all remaining 85 countries were used for model training but not Kazakhstan. For sensitivity assessment related to the choice of reference population, a post-Soviet model (Russia, Ukraine, Belarus, Lithuania, Latvia, Estonia)[7,5] was also fitted, drawn from countries that share the legacy of the Soviet healthcare system. A data-driven structural-peer group was further explored using k-means clustering of standardized country-level macroeconomic and consumption indicators ($k = 6$); however, Kazakhstan emerged as a singleton cluster, with its closest neighbours being mostly post-Soviet and Eastern European middle-income countries. As no distinct peer group could be formed, no structural-peer comparison was conducted, and inference relied on the global LOCO model.

2.4 Model Development

The following algorithms were experimented with: gradient-boosted models (LightGBM, CatBoost) and ensemble/interpretable baselines (Random Forest, Logistic Regression). For the gradient-boosted models, positive-class weight (`scale_pos_weight`) was used to handle class imbalance (resistant/susceptible $\approx$ 1:4), while balanced class weights were applied for Logistic Regression and Random Forest. Owing to computational constraints, hyperparameters were tuned once on the full global dataset rather than separately within each LOCO fold. However, this is not a major concern because no single country accounts for more than 5 per cent of the total 85 training countries.

2.5 Evaluation Methods

2.5.1 General Evaluation Metrics

The following metrics were used to assess the models' performance: AUC (Area Under the Curve), AUPRC (Area Under the Precision-Recall Curve), recall, F1 score, calibration-based metrics (Brier score and BSS: Brier Skill Score), and ECE (Expected Calibration Error). For this reason, recall and AUPRC were selected as co-primary metrics, as resistant infections may have clinical implications because falsely treated resistant infections may lead to further spread of resistance, while falsely considered "susceptible" infections do not[18].

2.5.2 Calibration Methods

The degree of calibration was evaluated using ECE and BSS. Isotonic regression was performed on the validation cohort (2019–2020), while the test set and Kazakhstan inference sets were calibrated by isotonic regression. Calibration data were mainly gathered from high-income countries, which might lead to domain shift when transferring the calibration data to Kazakhstan's dataset; this can be mitigated by the post-Soviet cluster, considering the similarities in the macro-feature space.

2.5.3 Pairwise Model Comparison Method

Six paired comparisons of AUCs were performed between the four candidate models using the DeLong test[19]. The main procedure was Benjamini-Hochberg FDR correction: a collection of 6 raw p-values (one for each test) was assembled, and each p-value was corrected for FDR using the standard step-up procedure (p-values were ranked, and each was compared to the modified critical value $0.05 \times (i/6)$, where i is the rank). At $\alpha = 0.05$, significance was determined based on the corrected p-value. For a full picture, Bonferroni-corrected p-values (typically a conservative correction with a large n) are also reported in Supplementary Table S6.

2.5.4 Validation Methods

For geographic generalization testing, we utilized LOCO cross-validation (with early stopping during the years 2019–2020) and four-country hold-out testing (in which each of the four countries had no data in the training set and no data from the held-out set were used for early stopping, and each country was held out separately to test the models with its 100% dataset).

A temporal robustness evaluation technique was adopted: rolling-origin (expanding-window) cross-validation. The training set consisted of all data up to the year specified, and the validation set consisted of data from the following year. Thus, validation data were always in the future relative to the training data ($\leq$2016 $\rightarrow$ 2017, $\leq$2017 $\rightarrow$ 2018, $\leq$2018 $\rightarrow$ 2019, $\leq$2019 $\rightarrow$ 2020). The leakage-safe discipline of the main pipeline was followed: parameters (imputation, `scale_pos_weight`, isotonic calibration) were re-derived using the training data from each fold, and early stopping was applied to an inner subset of the training years, not the validation year. Discrimination (AUC, AUPRC) was computed from raw scores, while calibration-dependent measures were computed from probabilities after isotonic calibration.

Spearman's rank correlation analysis was used to examine the trend in predicted resistance in Kazakhstan, with false-discovery rate (FDR) correction based on the Benjamini-Hochberg procedure at the organism level (2017–2023) for the six pathogens.

2.5.5 Feature Analysis Methods

Feature importance was computed for the CatBoost model using the PredictionValuesChange metric, while SHAP values were computed on the stratified test set (n = 5,000) using TreeExplainer[20]. Note that the two measures can yield different rankings when there is temporal correlation among features: PredictionValuesChange measures the difference across all predictions, and SHAP measures the marginal contribution of each feature.

2.5.6 Face Validity Method

The primary model's calibrated resistance probabilities were compared with the resistance rates reported in published clinical studies in Kazakhstan to check face validity. Antibiotic–pathogen combinations were chosen that were reported with explicit resistance percentages for the pathogen with clear endpoints, and combinations of these antibiotic–pathogen relations were present in the inference grid. For each of the combinations, the predicted value was the average calibrated probability for the years in question. There was an expected downward bias because the model estimates national-level resistance while the studies reported higher-acuity hospitalized subpopulations, which necessitated a directional interpretation of the comparisons with an absolute deviation of more than 25 percentage points recorded as a substantive difference. Since there were a limited number of similar combinations ($n = 4$), a formal statistical test was not used.

2.6 Data Availability

The data and code reported in this article are made available on Zenodo (`https://doi.org/10.5281/zenodo.20727165`). The deposit contains a single reproducibility notebook implementing the full pipeline, together with the supplementary reference tables (`reference_tables.pdf`: organism-specific MIC breakpoints, antibiotic-to-class and Kazakhstan antibiotic name mappings, and per-model hyperparameters) and the complete antibiotic–organism–year resistance-probability grid for Kazakhstan (`kz_predictions.csv`). The global isolate-level training data (ATLAS: Pfizer; SIDERO-WT: Shionogi; KEYSTONE) are governed by Data

Use Agreements with Vivli and cannot be redistributed freely; reproducing the full training pipeline therefore requires a separate data-access application through Vivli (`https://vivli.org`).

2.7 Ethics

The ethical review was carried out by Assistant Professor Dauren Ayazbayev, PhD, from SDU University – Department of Information Systems. The study did not involve any human subjects, nor did it involve the collection of any biological specimens, access to identifiable patient data, or, if such data were used, the keeping of any identifiable data that could allow identification of patients in the course of the study; therefore, it was determined exempt from IRB review.

Results

3.1 Data Overview

The final dataset comprised 11,275,172 susceptibility results indicating whether an organism was resistant or susceptible to a given antibiotic, spanning 85 countries over 19 years (2004–2023). Resistant results accounted for 19.8% (2,236,071 resistant samples out of 11,275,172). Under the temporal split, there were 7,203,908 training records ($\leq$2018, 18.9% resistant), 1,601,179 validation records (2019–2020, 21.6% resistant), and 2,470,085 hold-out test records (>2020, 21.4% resistant).

3.2 Model Performance

On the hold-out dataset (>2020), LightGBM attained the highest AUC (0.8512), and CatBoost the highest AUPRC (0.6496) and Recall@0.2 (0.7970). The selected final model was CatBoost, which achieved the pre-specified Recall score and AUPRC (Section 2.5.1). CatBoost's test values for AUC, AUPRC, and F1@0.2 were 0.8490, 0.6496, and 0.5612, respectively (Table 2). In the pairwise comparisons, after Benjamini–Hochberg correction of p-values from the DeLong test, all six pairwise AUC differences were statistically significant; however, the CatBoost–LightGBM difference was significant but negligible in magnitude (ΔAUC = 0.0022). Both gradient-boosted models were significantly better than Random Forest and Logistic Regression ($p_{\mathrm{BH}} < 0.001$). Isotonic regression improved ECE for CatBoost from 0.1780 to 0.0120, raised the Brier skill score from -0.0017 (pre-calibration) to 0.3131, and did not change AUC (Table 2).

Four-fold rolling-origin temporal cross-validation yielded a mean AUC of 0.870 ± 0.016, mean AUPRC of 0.681 ± 0.006, and mean F1@0.2 of 0.585 ± 0.006. The cross-validation AUC modestly exceeds the independent hold-out estimate (0.870 vs 0.849). The confusion matrix is presented in Figure 2a.

Table 2: Model performance and calibration on the hold-out test set (>2020). Metrics are computed on isotonic-calibrated probabilities. @0.2 and @0.5 denote the classification threshold applied to the calibrated probability: 0.20 is the clinical actionability threshold (primary operating point); 0.50 is the conventional reference threshold.

Model	AUC	AUPRC	Brier	BSS	ECE	Recall@0.2	Prec@0.2	F1@0.2	Recall@0.5	Prec@0.5	F1@0.5
CatBoost	0.8490	0.6496	0.1155	0.3131	0.0120	0.7970	0.4330	0.5612	0.4098	0.7222	0.5229
LightGBM	0.8512	0.6481	0.1153	0.3142	0.0122	0.7822	0.4447	0.5671	0.4139	0.7082	0.5225
Random Forest	0.8199	0.5771	0.1255	0.2537	0.0076	0.7376	0.4264	0.5404	0.3255	0.7032	0.4450
Logistic Regression	0.7940	0.5269	0.1335	0.2061	0.0135	0.7411	0.3962	0.5163	0.2763	0.6523	0.3882

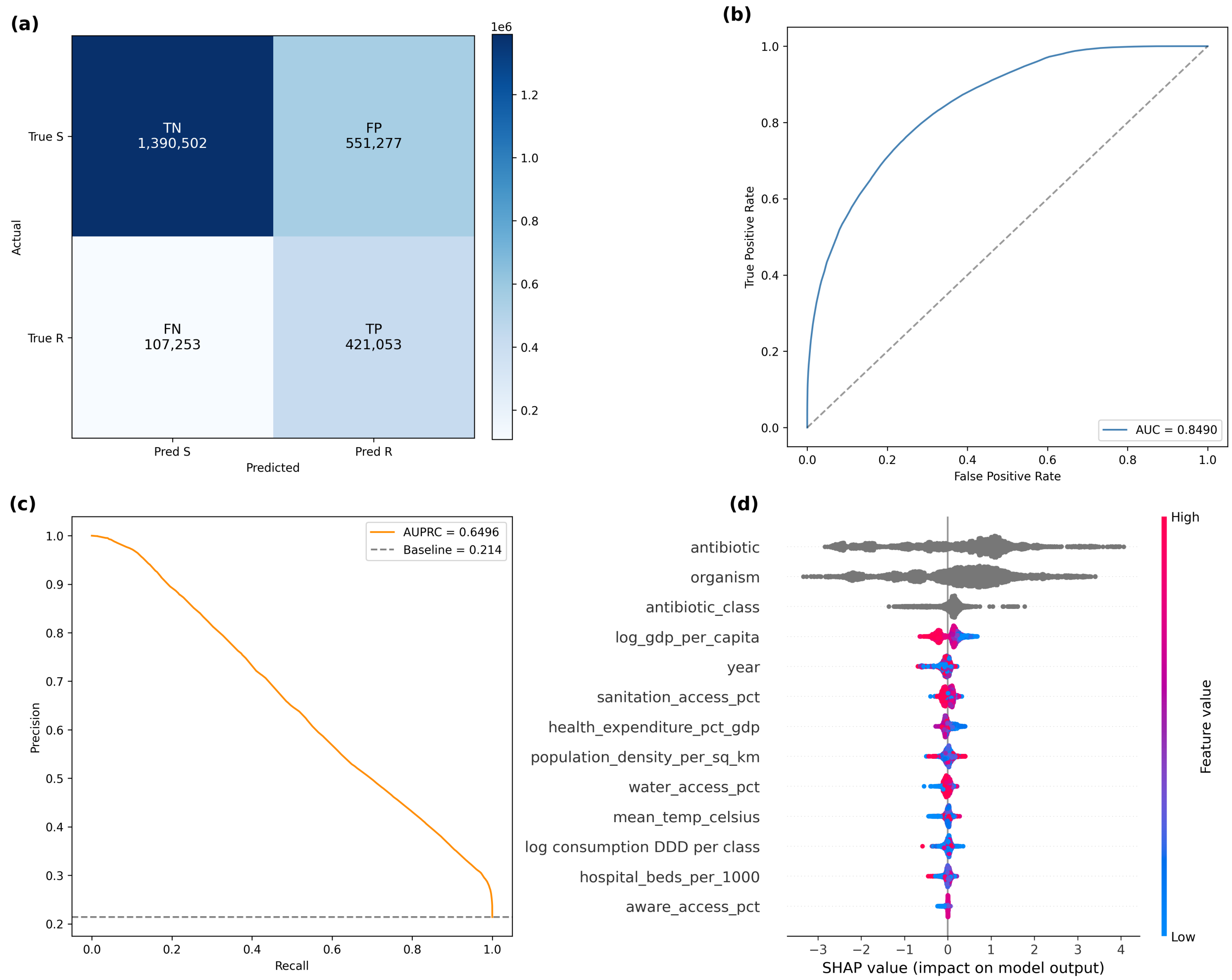

Figure 2: Global hold-out evaluation of the CatBoost classifier (test isolates ≥2021; n = 2,470,085). (a) Confusion matrix at the 0.20 threshold. (b) ROC curve (AUC = 0.849). (c) Precision–recall curve (AUPRC = 0.650; baseline = 0.214). (d) SHAP beeswarm (n = 5,000) ranked by mean |SHAP|.

3.3 Feature Analysis

Organism (35.9%) and antibiotic (27.3%) were the most important features, with the three categorical features (organism, antibiotic, and antibiotic class, 11.8%) contributing 75.0% of total predictive importance. The most important numeric measures are year (4.5%), sanitation access (4.4%), log GDP per capita (3.1%), and per-class antibiotic consumption (log consumption DDD per class, 2.9%). None of the macroeconomic features were individually sufficient for the test AUC, as their removal did not impact the score by more than 0.0043, and their elimination (retaining only numerical features) brought the discriminative score to 0.618 from 0.848. As seen from the SHAP analysis in Figure 2d, on the one hand log consumption DDD was concentrated around zero and had a small positive trend, and on the other hand, higher values of log GDP per capita and health expenditure were correlated with negative SHAP values. The difference between the two rankings of the temporal features, year, results from the difference between the two methods described in Section 2.5.5.

3.4 Validation Results

The results of cluster-based and LOCO validations are included in Table 3. Although the post-Soviet model uses only 2.5% of the training data, it achieved the highest AUC (0.8979) with only a marginal difference in its Brier score after isotonic calibration (0.1389 to 0.1186). The global LOCO model (all countries except Kazakhstan) achieved an AUC of 0.8812 and an AUPRC of 0.7050. For country hold-out testing (Table 4), the calibrated Brier score ranged from lowest in France (0.0808) to highest in India (0.2135), suggesting that the transferability of calibration remained variable across countries.

With the global LOCO model used as the main model, the overall mean calibrated resistance probability across the 480 inferences for Kazakhstan was 0.338, with 66.9% above the 20% clinical actionability threshold. *Acinetobacter baumannii* (mean resistance prediction 0.55) was the most resistant, followed by *Klebsiella pneumoniae* and *Enterococcus faecium* (∼0.43 mean resistance prediction).

Table 3: Performance of post-Soviet and global LOCO model including Kazakhstan inference statistics.

Group	n_countries	n_train	n_val	val_AUC	val_AUPRC	Brier_raw	Brier_cal	kz_mean_cal	kz_%≥0.2
post soviet	6	179,237	60,583	0.8979	0.8073	0.1389	0.1186	0.3996	72.5
LOCO (KZ)	85	7,203,908	1,601,179	0.8812	0.7050	0.1550	0.1059	0.3380	66.2

Table 4: Country hold-out evaluation for Russia, Turkey, France, and India.

Country	n_test	Prevalence	AUC_cal	AUPRC_cal	Recall@0.2_cal	Prec@0.2_cal	F1@0.2_cal	Brier_cal
Russia	169,418	0.2772	0.8856	0.7418	0.7777	0.6057	0.6810	0.1341
Turkey	182,604	0.2877	0.8454	0.6922	0.8233	0.5188	0.6365	0.1471
France	640,323	0.1479	0.8879	0.6442	0.8020	0.3920	0.5267	0.0808
India	205,815	0.4411	0.8074	0.7592	0.7343	0.6606	0.6955	0.2135

3.5 Kazakhstan Resistance Inference

Table 5: Predicted probabilities of resistance (2017–2023) with associated 95% bootstrap CIs (1,000 resamples) for six WHO-priority pathogens for Kazakhstan, based on the global LOCO model. Spearman trend statistics are ρ, p(BH), and p(Bonf).

Organism	Mean (2017–2023)	95% CI	2017	2023	Trend	rho	p (BH)	p (Bonf)
A. baumannii	0.551	0.508–0.591	0.562	0.662	rising	0.786	0.0489	0.2174
K. pneumoniae	0.434	0.383–0.486	0.399	0.561	rising	0.536	0.2152	>0.99
E. faecium	0.423	0.365–0.481	0.376	0.609	rising	0.786	0.0489	0.2174
P. aeruginosa	0.269	0.242–0.295	0.291	0.348	rising	0.775	0.0489	0.2446
E. coli	0.267	0.226–0.307	0.201	0.402	rising	0.786	0.0489	0.2174
S. aureus	0.097	0.080–0.115	0.111	0.141	rising	0.786	0.0489	0.2174

The mean calibrated RP for each of Kazakhstan's six WHO-priority pathogens (2017–2023) is reported in Table 5, with 95% bootstrap (1,000 resamples) CIs to show uncertainty.

Three pathogens had high mean predicted resistance (PMR) values, with *Acinetobacter baumannii* (0.551) being the most resistant, followed by *Klebsiella pneumoniae* (0.434) and *Enterococcus faecium* (0.423). *Staphylococcus aureus* had the lowest (0.097; 95% CI 0.080–0.115). None of the organisms exhibited a decrease in predicted probability during the study period from 2017 to 2023. A Spearman rank correlation with Benjamini–Hochberg FDR correction was performed on the annual mean series for the 6 pathogens. Of these organisms, five showed positive trends that passed FDR ($p_{\mathrm{BH}} < 0.049$) but did not meet the Bonferroni correction; these are considered hypothesis-generating and should not be considered conclusive. *K. pneumoniae* showed no significant trend ($\rho = 0.536$, $p_{\mathrm{BH}} = 0.215$).

By antibiotic class, predicted resistance was highest for penicillins, driven almost entirely by ampicillin and most pronounced in *Klebsiella pneumoniae* (0.99) and *Enterococcus faecium* (0.89) — consistent with the intrinsic and near-universal ampicillin resistance established for these organisms. Among the remaining classes, cephalosporins were highest for *A. baumannii* (0.73) and *K. pneumoniae* (0.54), and fluoroquinolones for *E. faecium* (0.56).

Discussion

4.1 Key Findings

The hypothesis that the determinants of AMR are sufficiently universal to allow forecasting at a cross-country level prior to gathering local data was tested. The model performed well in validation and supported—though did not prove—the hypothesis, because the model had a global hold-out AUC score of 0.849 and a LOCO validation AUC score of 0.881 without using any Kazakhstani training set samples. There is a lack of validation samples—isolates—from Kazakhstan against which performance can be directly measured. The biggest predictors were organism and antibiotic identity (75% of the total importance), with the removal of this information causing a decrease in AUC from 0.848 to 0.618, while no macroeconomic group removed led to a change in AUC of more than 0.004. Accordingly, the ranking of resistance is mainly dependent on the organism–antibiotic combination and agrees well with MacFadden et al.[6]. The only pathway through which country-level variation arises in the Kazakhstan inference is in the macroeconomic and consumption features, with their overall contribution to the global discrimination being unimportant. A model for Kazakhstan identified *A. baumannii* as the most worrisome pathogen (mean 0.55), followed by *K. pneumoniae* and *E. faecium* (means of ∼0.43); higher GDP and health expenditure were correlated with lower predicted resistance, consistent with Laxminarayan et al.[4]. Overall, 66.9% of inferences fell above the 20% actionability threshold.

4.2 Implications

Most countries without isolate-level AMR surveillance possess countrywide data and antibiotic consumption knowledge; for those that lack such resources, a viable precedent exists to help prioritise scarce AMR surveillance resources in the interim. A health system may, however, obtain a ranked, organism-specific risk rating that will help determine pathogen–antibiotic combinations that need to be initially confirmed. This decoupling of resistance levels (intrinsic resistance and absolute resistance) explains why resistance ranking can be transferred even without calibration: discrimination alone can be transferred across borders without calibration.

4.3 Model Selection

Selection was done on a clinical basis, not a statistical basis. Statistically, CatBoost and LightGBM were not significantly different with respect to discrimination and calibration (AUC 0.849 vs 0.851, calibrated BSS 0.313 vs 0.314); the DeLong test rendered the 0.002 AUC difference statistically significant, reflecting sensitivity at multi-million-row scale rather than a clinically meaningful gap. Selection followed the pre-specified recall criterion, on which CatBoost was higher (0.797 vs 0.782); because the tool prioritizes which combinations to test, maximizing recall (catching resistant cases) matters more than minimizing false alarms. ECE was reduced from 0.178 to 0.012 with isotonic calibration, and the 20% limit is in line with previous stewardship efforts[21].

4.4 Applications and Potential Research Directions

Most previous AMR prediction algorithms rely on genome sequencing[22] or on local surveillance[6]; thus, they may not be available in settings lacking surveillance or sequencing. In this method, only macro-proxy data are used to predict resistance rates for all six priority pathogens using no locally isolated pathogens. The AUC achieved by its hold-out data of 0.849 is very close to per-drug genomic AUC models (0.83–0.94)[25] and species-specific MALDI-TOF models (0.81–0.85)[24] and better than EHR machine-learning-based patient-specific AUC models (0.80)[26] — exactly as expected when launching without local data, which is the scenario where a cross-country applied XGBoost model achieved an AUC of 0.96[23] on this same database.

4.5 Kazakhstan Inferences and Face Validity

Broken down by antibiotic class, predicted resistance was dominated by penicillins, reflecting the intrinsic ampicillin resistance of *K. pneumoniae* and *E. faecium* — a pattern the model recovered without any local data, supporting its face validity. Among the remaining classes, cephalosporins were highest for *A. baumannii* and *K. pneumoniae*, and fluoroquinolones for *E. faecium*. With four combinations of *A. baumannii* from a single cohort (Karaganda; Table 6), absolute probabilities range from 15 to 26 percentage points off, allowing the model to confidently identify *A. baumannii* as high-risk (all predicted >70%), yet demonstrating the difference in absolute probabilities in a higher-acuity subpopulation of hospitalized patients. The comparison is suggestive, which is not the same as confirmation.

Table 6: Face validity comparison of predicted versus observed *A. baumannii* resistance rates from Strochkov et al.[27].

Organism	Antibiotic	Observed_%	Predicted_%	Difference	Direction	Year_range	Setting	Region
A. baumannii	Gentamicin	88.89	70.4	−18.5	Under 10–25pp	2022–2023	Pneumonia inpatient	Karaganda
A. baumannii	Ceftriaxone	65.38	84.1	18.7	Over 10–25pp	2022–2023	Pneumonia inpatient	Karaganda
A. baumannii	Ciprofloxacin	65.38	80.6	15.2	Over 10–25pp	2022–2023	Pneumonia inpatient	Karaganda
A. baumannii	Fluoroquinolones (class)	100.00	74.2	−25.8	Under-predicts >25pp	2022–2023	Pneumonia inpatient	Karaganda

4.6 Limitations

The chief limitation is that the model was not trained or tested using isolate-level information from Kazakhstan, and the results reported for the country are preliminary and will need to be tested using future data. Some partial resistance was not represented, as the partially resistant isolates were not included and some antimicrobial medicines with wider usage in the area were not represented. The data in this section are mostly drawn from high-income groups, and so absolute probabilities should be treated with caution rather than as validated accuracy; the confidence intervals in Table 5 are based on prediction consistency rather than on an absolute measure of the accuracy of the predictions. The upward temporal trend predictions, ranked as hypothesis-generating signals, are based on a limited number of annual observations (7 per organism) and were significant only under Benjamini–Hochberg correction ($p_{\mathrm{BH}} = 0.049$) but not Bonferroni correction. Finally, hyperparameter tuning was performed on the entire dataset rather than on each fold, which may overestimate LOCO-AUC, but no single country exceeds 5% of the data in any fold[28].

4.7 Future Directions

An immediate validation (100–200 local isolates of high-risk species (*A. baumannii*, *K. pneumoniae*, and *E. faecium*)) will be performed to compare with model estimates, undertaken with a narrow band of ±10 per cent compared with the wider band of ±25 per cent used for the retrospective validation. The framework can be extended, not only to additional sectors that have good national statistical coverage such as much of Central Asia, sub-Saharan Africa, and Southeast Asia, but also to other areas more broadly.

Conclusion

Following a cross-country methodology based on non-genetic information, we estimated the probability of antimicrobial resistance (AMR) in Kazakhstan, where we lacked direct information from local isolates, relying only on probabilities derived from global data on AM resistance, antibiotic use, and macro-economics. The CatBoost model performed best, with 11.3 million susceptibility results from 1.08 million isolates from 85 countries in the study, achieving an AUC of 0.849, an AUPRC of 0.650, and a recall of 0.797, identifying *A. baumannii*, *K. pneumoniae*, and *E. faecium* as the top-risk pathogens in Kazakhstan. The framework does not replace laboratory surveillance but is a hypothesis-generating screening tool that prioritises which combinations to validate first. If surveillance is lacking, prediction need not be: global information can serve as a provisional map of resistance risk to determine where to collect the first local samples.

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